

Agency-Dependent Fairness Evaluation

A Role-Counterfactual Audit of Political Fairness in Language Models

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Abstract

Agency-Dependent Fairness Evaluation (ADFE) asks whether a language model’s political-fairness behavior depends on the *civic role* it is assigned. The method is a role-counterfactual audit: hold the policy task, topic, source packet, audience, and risk level fixed; vary the assigned role over an agency ladder; and score outputs on a six-part fairness signature with a held-out judge that is separately validated against human-labeled datasets. After hardening the run pipeline against silent artifact failures, duplicate rows, self-judging, prompt-tuning contamination, and circular outcomes, the repaired clean local run contains $n = 2,100$ scored outputs across five small local models. The mixed-effects primary test finds positive role-agency slopes on all six dimensions. This is evidence of role sensitivity, but not a simple confirmation of the original theory: invariant floor dimensions, factuality (**E**) and non-manipulation (**M**), also rise with agency, suggesting a broad role-elaboration or answer-quality effect alongside the predicted rise in curation and deliberative dimensions. The most operationally concrete small-model risk is viewpoint-asymmetric refusal. On the frontier Grok run ($n = 630$), refusal asymmetry vanishes, but the small local judge saturates on several dimensions, showing that judge capability must match the audited population.

1 Introduction

Most political-fairness audits ask whether a model is biased. That question is under-specified. A personal assistant drafting a lawful partisan argument, a campaign aide preparing messaging, a government information service explaining a procedure, and a civic news provider curating what is salient do not have identical obligations. Fairness depends not only on topic and viewpoint, but on the *role* the system has been asked to occupy.

ADFE makes that claim testable. Roles are experimental treatments. Each output is scored on a six-dimensional fairness signature, and the primary outcome is a regression of each dimension on a continuous role-agency level. The theory predicts that dimensions tied to editorial discretion and public-facing curation should rise with agency, while invariant floors such as factuality and non-manipulation should stay high regardless of role.

2 Model and measurements

2.1 Six fairness dimensions

Each output is judged on raw 0–4 scores, normalized to $[0, 1]$:

- U: User-agency fidelity.
- E: Epistemic integrity and source-grounded factuality.

- V: Viewpoint symmetry across analogous lawful viewpoints.
- C: Curation accountability: salience, sourcing, and uncertainty separation.
- D: Deliberative standing for disagreement and minority views.
- M: Non-manipulation and viewpoint-neutral refusal integrity.

2.2 Roles as an agency ladder

Seven roles are ordered by agency level α : assistant and advocate (0.20), campaign aide (0.35), researcher (0.45), government information service (0.50), mediator (0.60), and civic news provider (0.75).

2.3 Primary test

For each dimension d , the primary model is:

$$y_{i,d} = \beta_{0,d} + \beta_{1,d}\alpha_{r(i)} + b_{m(i),d} + \epsilon_{i,d}, \quad b_{m,d} \sim \mathcal{N}(0, \sigma_{m,d}^2).$$

Here $\alpha_{r(i)}$ is the agency level of the assigned role, and $b_{m(i),d}$ is a random intercept for the audited model. Paired viewpoint gaps are computed as:

$$\Delta_{\text{refusal}}(p, \bar{p}, m, r) = |\mathbb{K}[\text{refuse}(p, m, r)] - \mathbb{K}[\text{refuse}(\bar{p}, m, r)]|,$$

$$\Delta_{\text{quality}}(p, \bar{p}, m, r) = \frac{1}{4} \sum_{d \in \{U, E, V, M\}} |y_{p,d} - y_{\bar{p},d}|.$$

3 Run integrity and methodology hardening

The pipeline was hardened before the current results were published:

- Generation failures are written only to `generation_errors.jsonl`; failed generations are retryable and are not silently scored as zero.
- `audit-run` fails runs with duplicate item IDs, errored generation rows, orphan scores, incomplete full-factorial artifacts, contaminated publication runs, or missing frozen configs.
- `repair-run` backs up artifacts, drops bad generation/score rows, deduplicates by `item_id`, and rewrites analysis.
- Frozen runs resume from `frozen_config.yml`; config/design hashes prevent accidental axis drift.
- The judge is held out from the audited model set and validated per construct.

4 Experimental setup

The clean local run uses 30 prompts, 6 analogous viewpoint pairs, 6 topics, 7 role cards, 2 agency modes, and 5 small local models: `llama3.2:3b`, `llama3.2:1b`, `phi3:mini`, `gemma3:1b`, and `deepseek-r1:1.5b`. It contains 2,100 successful generations and 2,100 scores.

The secondary frontier run uses three xAI Grok variants and contains 630 successful generations and 630 scores. Both runs pass `audit-run --expect-full`.

5 Judge validation

Dimension	Dataset	κ	Accuracy	Interpretation
M safety/refusal	XSTest ($n = 450$)	0.7753	0.8911	strong enough for refusal analysis
E factuality	TruthfulQA ($n = 1,580$)	0.5000	0.7500	moderate; use caution
V viewpoint	BABE ($n = 907$)	0.2966	0.6362	weak; conclusions are tentative

Table 1: Judge validation is dimension-specific. A single global judge score would hide important blind spots.

6 Clean local results

6.1 Primary agency-gradient test

Dimension	Slope on α	95% CI	p	Result
U	+0.0965	[0.0420, 0.1511]	0.0005	positive
E	+0.0587	[0.0027, 0.1148]	0.0401	positive floor movement
V	+0.0906	[0.0365, 0.1447]	0.0010	positive
C	+0.0769	[0.0215, 0.1324]	0.0066	positive
D	+0.1157	[0.0604, 0.1709]	< 0.0001	strongest slope
M	+0.0622	[0.0188, 0.1055]	0.0049	positive floor movement

Table 2: All six dimensions rise with assigned role agency in the repaired clean run. This supports role sensitivity, but because the invariant floors E and M also rise, it is not a clean separation between discretionary duties and fixed floors.

6.2 Role-band predictions

The pre-specified role-interval checks are mixed: 16 role $\times$ dimension predictions are supported, 17 are violated, and 9 are inconclusive. This is why the headline is not simply "the theory is confirmed." The data show role-conditioned movement, but the exact normative bands require revision or human calibration.

6.3 Model heterogeneity and refusal asymmetry

7 Frontier results

The Grok run has a refusal rate of 0.0 and no viewpoint refusal-parity gap. However, the local qwen3:8b judge saturates on several dimensions. Curation (C), factuality (E), and non-manipulation (M) have insufficient variance for a meaningful gradient estimate. User fidelity (U) is near-ceiling and not significant (+0.0026, $p = 0.2118$). Deliberative standing (D) and viewpoint symmetry (V) rise with agency (+0.1103 and +0.1180, respectively), but the viewpoint gate is the judge’s weakest validated construct. The honest frontier conclusion is therefore methodological: capable models require a capable, separately validated judge.

Model	Refusal rate	Refusal-parity gap	Viewpoint-quality gap
deepseek-r1:1.5b	0.3286	0.2738	0.1775
llama3.2:1b	0.1310	0.2024	0.2143
llama3.2:3b	0.0476	0.2381	0.2303
phi3:mini	0.0167	0.0476	0.0859
gemma3:1b	0.0000	0.0000	0.0751

Table 3: Small models differ sharply. The most concrete deployment risk is not total refusal rate alone, but asymmetric refusal across paired lawful viewpoints.

8 Deployment ramifications

ADFE is best understood as a deployment test pattern:

- Treat role prompts as versioned policy configuration, not incidental copy.
- Include role-counterfactual and viewpoint-paired tests in release gates.
- Report refusal parity across lawful mirrors, not only aggregate refusal rate.
- Validate evaluators per construct and watch for saturation.
- Preserve logs, frozen configs, score checks, and artifact audits.
- Route high-impact civic roles to human review and escalation.

9 Legal and governance context

This report is not legal advice. Still, role-counterfactual evidence is relevant to governance because current AI risk regimes already emphasize deployment context, intended purpose, monitoring, and accountability. NIST’s AI Risk Management Framework organizes governance around Govern, Map, Measure, and Manage. The EU AI Act is risk-based and assigns obligations to developers and deployers depending on system use and classification. U.S. agencies have emphasized that existing civil-rights, consumer-protection, credit, employment, and competition laws apply to automated systems; sectoral examples include EEOC guidance for AI in employment and CFPB requirements for specific adverse-action reasons when complex algorithms are used in credit decisions.

10 Limitations

The prompt bank is small and U.S.-policy focused. Scores are LLM-judge measurements, not in-domain human ratings. The judge is strong on refusal, moderate on factuality, weak on viewpoint symmetry, and too small for reliable frontier scoring. The role-band intervals are researcher-specified predictions and should be treated as hypotheses, not ground truth. Frontier results cover one model provider and should not be generalized across the frontier-model ecosystem.

11 Conclusion

The hardened ADFE run shows that assigned civic role can move political-fairness behavior in measurable ways. The repaired clean local result is more interesting than a simple confirmation or falsification: all dimensions rise with agency, including dimensions that were expected to remain flat.

That makes the next research question sharper. Is the model learning a normatively meaningful role distinction, or is it simply producing more careful, elaborate answers for high-agency civic roles? The current harness is now able to ask that question without silently accepting bad artifacts, self-judging loops, or stale run configurations.

References

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